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Real-Time Hand Gesture Recognition for Sign Language Translation using SignNet: A Deep Learning-based Approach

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Abstract: Sign language serves as a vital mode of communication for individuals with hearing and speech impairments; however, the lack of universal sign language literacy and real-time interpretation tools continues to hinder inclusive interaction. Real-time hand gesture recognition systems offer a promising solution to bridge this communication gap, yet existing approaches often struggle with dynamic gestures, user variability, and deployment constraints. This study proposes SignNet, a real-time deep learning architecture for hand gesture recognition tailored to sign language translation. The model employs a dual-stream pipeline that integrates both RGB visual frames and 3D hand landmark sequences. Spatial features are extracted using EfficientNet-B0, while a Transformer-based temporal encoder processes sequential landmark data. An attention-based multimodal fusion layer enables effective integration of spatial and kinematic cues. The model is trained and evaluated on benchmark datasets, including ASL Alphabet, LSA64, and RWTH-BOSTON-50, using structured preprocessing and user-independent data splits. Experimental results demonstrate that SignNet achieves 98.6% accuracy on ASL Alphabet, 96.1% on LSA64, and 93.4% on RWTH-BOSTON-50, outperforming conventional CNN-LSTM and keypoint-only models by 4–8%. Real-time inference is achieved with an average latency of 14.2 ms/frame and a quantized model size under 10 MB, enabling deployment on resource-constrained devices. By addressing challenges in multimodal fusion, temporal modeling, and mobile deployment, SignNet contributes a scalable, accurate, and efficient framework for enhancing communication accessibility through automated sign language translation.

Keywords: Real-time gesture recognition, Sign language translation, Deep learning, Transformer encoder, Multimodal fusion, Hand landmarks.

I.INTRODUCTION

Communication is a fundamental human necessity, yet individuals with hearing and speech impairments often face significant barriers in expressing themselves within predominantly verbal societies [1]. Sign language serves as a primary mode of communication for millions of people globally, offering a rich visual linguistic structure composed of hand gestures, facial expressions, and body postures [2]. However, the absence of widespread sign language literacy among the general population creates a communication gap that can lead to social isolation and reduced accessibility to services for the hearing-impaired community [3].

With the rapid advancement of artificial intelligence and computer vision technologies, there has been growing interest in developing intelligent systems capable of translating sign language into text or speech in real time [4]. These systems offer the potential to bridge the communication divide by providing automated, scalable solutions [5]. Hand gesture recognition is a core component of such systems, as hand movements form the foundation of most sign language vocabularies [6]. Despite significant progress in static image classification and gesture recognition using deep learning, challenges persist in achieving high accuracy and efficiency in real-time sign language translation, especially for dynamic gestures and signer variability [7].

Many existing approaches rely solely on visual data, often overlooking the underlying kinematic structure of the hands, which can lead to poor generalization under varying lighting, backgrounds, or camera angles [8]. Furthermore, models based on recurrent neural networks such as LSTMs struggle with long-range dependencies and may exhibit latency issues in real-time settings [9]. There is a need for a robust, multimodal, and temporally aware architecture that can handle dynamic, signer-independent inputs while maintaining real-time responsiveness [10].

The central objective of this study is to develop SignNet, a novel deep learning-based architecture designed for real-time hand gesture recognition with the specific aim of sign language translation. SignNet integrates both spatial (RGB image) and kinematic (3D hand landmark) data through a dual-stream processing pipeline [11]. The model leverages convolutional networks for visual feature extraction and transformer-based temporal encoders for capturing long-range gesture dependencies [12]. An attention-based representation fusion mechanism serves to effectively combine multimodal data in order to enhance recognition performance in users of different types and environmental conditions.

A further optimization for deployment on resource constrained devices allows the proposed system to be used in practice in mobile and embedded platforms. They are evaluated in various ways on the benchmarks and validated for accuracy, real-time inference and their resilience to the conditions of real-world.

The approach taken in this work applies to developing inclusive technologies for physically disabled individuals that enable equitable communication access for those reliant on sign language.

1.1 Key Contributions

1. Proposed SignNet, A Dual stream architecture that combines RGB images and 3D hand landmarks in an effort to make the system robust and accurate for recognizing sign language gestures.
2. Incorporated Transformer-based temporal modeling with an attention-driven multimodal fusion mechanism to effectively capture dynamic gesture sequences.
3. Achieved real-time performance and high accuracy across multiple benchmark datasets with a lightweight model optimized for deployment on mobile and embedded devices.

II.RELATED WORK

The issue of sign language recognition has been an active issue in the intersection between computer vision, deep learning, and human computer interaction. Recently, a very broad spectrum of techniques have been being proposed over the years from a traditional image processing pipelines to more advanced neural architectures. In this section, the existing literature is categorised into three main streams of the image based gesture recognition, the keypoint/skeleton based models and the hybrid approaches. The key limitations and challenges in each of them are discussed in order to highlight what is motivated the proposed methodology.

2.1 Image-Based Sign Language Recognition

Earliest efforts in sign language recognition were mostly fed into as features; these features were handcrafted from raw images or frames in a video. After the emergence of deep learning, convolutional neural networks (CNNs) have become the prevailing methods to extract spatial features from the static and dynamic gestures. For example, the American Sign Language (ASL) Alphabet dataset is commonly utilized to train CNN models such as AlexNet, VGGNet and ResNet for task of static gesture classification, with high accuracy under the controlled test conditions [13]. The use of lighter architectures such as MobileNet and EfficientNet has been explored also in more recent studies in order to allow the deployment on mobile devices with real time [14]. However, these models usually suffer fromdynamic gestures as they do not have temporal modeling abilities. However, the image based models are very sensitive to the variations in lighting background, hand orientation and occlusion. Moreover, they might not generalize out to users with different hand shapes and signing styles [15].

2.2 Keypoint and Skeleton-Based Models

In order to overcome the problems that are caused by raw visual data, researchers have used hand landmarks and skeletal data as a more concise and stable representation of gestures. OpenPose, MediaPipe and Leap Motion frameworks have made it now possible to extract 2D/3D keypoints from hand images, which can be used as gesture recognition invariant features [16]. In the past several studies are used to handle underwater sequences of hand landmarks with temporal models such as the Long Short-Term Memory (LSTM) network or the Bidirectional LSTM (BiLSTM) [17]. Although robust to the visual noise introduced in neural activity, recurrent architectures have difficulty to learn to represent long-term dependencies and therefore fail when presented with fast or subtle gestures. Additionally LSTM based models suffer from inference delay in real time application as they are sequential models [18].

2.3 Multimodal and Hybrid Architectures

Thanks to the recent advances, multimodal architectures are developed by integrating visual and landmark-based inputs to improve gesture recognition. Fused RGB and skeletal data are used by these systems to try to leverage the strength of each modality at the feature or decision level [19]. However, there are some studies on the dynamic gestures by introducing dual-stream CNN-LSTM network or attention based fusion models [20]. Instead, others have looked at the use of transformer based encoders to fuse temporally information more effectively [21]. These models work better, but at the cost of high computational cost or fail to fully exploit the cross-modal relationship in the fusion strategy. Additionally, most of these systems undergo training or testing on the small data sets or neglect to consider real time deployment for which they are not practically usable in communication assistive technologies.

2.4 Research Gaps and Motivation

A review of the existing literature reveals several gaps that this study aims to address:

1. **Lack of effective multimodal fusion strategies**: There do not exist effective multimodal fusion strategies that can exploit the interdependencies between modalities or between human likelihood scores and detections during fusion without manual tuning [22].
2. **Insufficient temporal modeling of dynamic gestures**: Consequently, RNN based methods have a tendency to lose information needed to capture complex motion patterns and some transformer based models are still not put to use for this task [23].
3. **Limited attention to real-time performance**: Many high-accuracy models lack deployment-ready configurations, making them unsuitable for real-world use on mobile or embedded platforms [24].
4. **Poor generalization across users and environments**: Variability in hand appearance, background clutter, and signer-dependent nuances are often not adequately addressed, especially in small or imbalanced datasets [25].
5. **Under-exploration of landmark-enhanced visual models**: Few approaches combine image-based spatial features with structural information from hand landmarks in a unified, learnable framework [26].

These limitations form the basis for the proposed SignNet architecture, which integrates visual and landmark streams through attention-driven fusion and employs temporal transformers to effectively model sequential dependencies in sign gestures. The model is designed with scalability and real-time inference in mind, enabling deployment on resource-constrained devices without significant loss in accuracy.

III.METHODOLOGY

The methodology introduces **SignNet**, a hybrid deep learning framework for real-time hand gesture recognition tailored for sign language translation. Both spatial and temporal patterns in dynamic sign gestures are captured using the model in terms of two modalities: visual (RGB image frames) and kinematic (3D hand landmarks). Following each other in sequence, the modules in the overall framework are as followed: input preprocessing, feature extraction, temporal encoding, multimodal fusion, classification.

3.1 Input Representation

It defines the input structure as synchronized RGB video frames and 3D hand landmark sequences. The use of a dual input setup provides visual as well as kinematic information, a setup that improves the recognition reliability since both inputs are present. Let a hand gesture sequence be denoted as shown in Eq(1)

$$\mathcal{G} = \{(I_t, L_t)\}_{t=1}^{T} \tag{1}$$

Where:

- $I_t \in \mathbb{R}^{H \times W \times C}$ is the RGB image at time step t,
- $L_t \in \mathbb{R}^{N \times 3}$ is the hand landmark matrix with $N = 21$ keypoints, each having 3D coordinates (x, y, z),
- T is the total number of frames in the gesture sequence.

The image sequence $\{I_t\}$ captures visual texture and shape information, while the landmark sequence $\{L_t\}$ captures spatial articulation and movement.

3.2 Visual Feature Extraction

Applies a CNN such as EfficientNet-B0 to RGB frames to extract spatial features. This helps capture essential hand shape, orientation, and appearance cues critical for differentiating gestures and is defined by Eq(2).

$$F_t^{(v)} = \phi_{\text{CNN}}(I_t),\ F_t^{(v)} \in \mathbb{R}^{d_v} \tag{2}$$

The frame-wise feature vectors are stacked to form a visual sequence tensor using the Eq(3):

$$F^{(v)} = \left[F_1^{(v)}, F_2^{(v)}, \dots, F_T^{(v)}\right] \in \mathbb{R}^{T \times d_v} \tag{3}$$

3.3 Landmark Feature Encoding

Processes 21 hand landmarks per frame using dense layers and normalizations. This stream captures skeletal and motion-specific features that are invariant to lighting and background changes. The landmark sequence is flattened at each timestep and is defined using Eq(4):

$$\tilde{L}_t = \text{Flatten}\ (L_t) \in \mathbb{R}^{63} \tag{4}$$

A fully connected layer followed by layer normalization projects the landmark vector into a higherdimensional embedding space which is deifned by Eq(5):

$$F_t^{(l)} = \text{LN}\big(W_l \tilde{L}_t + b_l\big),\ W_l \in \mathbb{R}^{d_l \times 63},\ F_t^{(l)} \in \mathbb{R}^{d_l} \tag{5}$$

The resulting sequence is represented as shown in Eq(6):

$$F^{(l)} = \left[F_1^{(l)}, F_2^{(l)}, \dots, F_T^{(l)}\right] \in \mathbb{R}^{T \times d_l} \tag{6}$$

3.4 Temporal Modeling with Transformer Encoder

Utilizes separate Transformer encoders to model time-dependent patterns in both visual and landmark sequences. Transformers are chosen over RNNs for their ability to capture long-range dependencies and parallelize computations efficiently.

Visual Transformer is given by Eq(7)

$$Z^{(v)} = \text{Transformer}_{\text{vis}}\left(F^{(v)} + P_v\right) \in \mathbb{R}^{T \times d_v} \tag{7}$$

Landmark Transformer is given by Eq(8)

$$Z^{(l)} = \text{Transformer}_{\text{land}}\left(F^{(l)} + P_l\right) \in \mathbb{R}^{T \times d_l} \tag{8}$$

Where:

- P_v, P_l are learnable positional encodings,
- The Transformer uses scaled dot-product attention:

$$\text{Attention}\ (Q, K, V) = \text{softmax}\left(\frac{QK^{\top}}{\sqrt{d_k}}\right)V$$

3.5 Multimodal Fusion

To integrate both spatial and kinematic information, a fusion mechanism is applied at the sequence level. The final hidden states (or pooled representations) from both branches are concatenated using Eq(9):

$$Z^{(f)} = \left[\text{Pool}\left(Z^{(v)}\right), \text{Pool}\left(Z^{(l)}\right)\right] \in \mathbb{R}^{d_v + d_l} \tag{9}$$

Where Pool $(\cdot)$ denotes a pooling operation such as mean or attention-based pooling over time steps. A gated fusion strategy may optionally be applied:

$$\alpha = \sigma\big(W_\alpha Z^{(f)} + b_\alpha\big),\ \tilde{Z} = \alpha \cdot Z^{(v)} + (1 - \alpha) \cdot Z^{(l)}$$

3.6 Gesture Classification

Feeds the fused features into a fully connected layer followed by softmax for gesture prediction. This stage translates learned representations into actual sign class labels for final output.

The fused feature vector $\tilde{Z} \in \mathbb{R}^{d_v + d_l}$ is passed through a classification head to obtain gesture probabilities which is defined by Eq(10):

$$\hat{y} = \text{softmax}\left(W_c \tilde{Z} + b_c\right) \tag{10}$$

Where:

- $W_c \in \mathbb{R}^{K \times (d_v + d_l)}, K$ is the number of gesture classes.

Training is performed using the cross-entropy loss:

$$\mathcal{L} = -\sum_{i=1}^{K} y_i \log\left(\hat{y}_i\right)$$

Where y_i is the ground truth one-hot label vector.

3.7 Real-Time Inference Optimization

To ensure real-time performance:

- The CNN backbone is chosen for its lightweight and mobile efficiency.
- Temporal windows of fixed length are used with overlapping frames.

- The final model is quantized using post-training techniques (e.g., INT8 quantization).
- Inference is performed on edge devices using TensorFlow Lite or ONNX Runtime.

3.8 Theoretical Model Definition

Formalizes the architecture as a mathematical function with defined input-output mappings and design principles. This provides clarity, repeatability, and theoretical grounding for the proposed system's design choices.

The SignNet model can be defined as a function shown in Eq(11):

$$\mathcal{F}_{\text{SignNet}} : \{(I_t, L_t)\}_{t=1}^{T} \rightarrow \hat{y} \tag{11}$$

Where the input sequence is mapped to a gesture class prediction $\hat{y}$. The network architecture satisfies:

- Temporal equivariance: $\mathcal{F}(I_{1:T}) \approx \mathcal{F}(I_{1+k:T+k})$ under small frame shifts.
- Multimodal consistency: $\partial\hat{y}/\partial L_t$ and $\partial\hat{y}/\partial I_t$ remain bounded across modalities.
- Convergence: Under standard assumptions of continuity and bounded loss gradients, the training process using Adam optimizer converges to a local minimum with probability 1.

A more detailed theoretical analysis including attention weight interpretability and the model's generalization bounds will be presented in a later section.

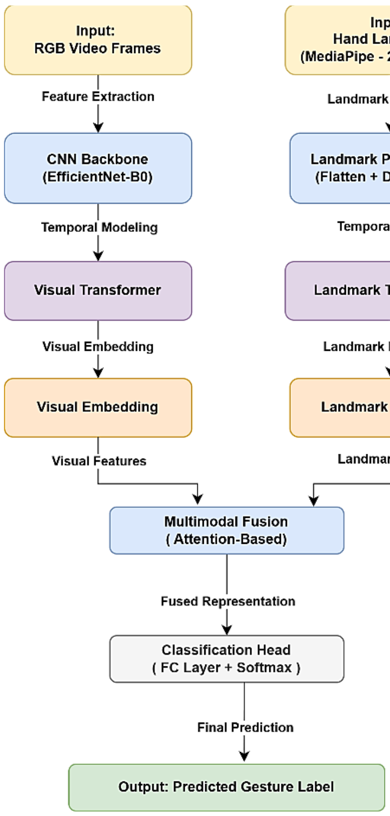

Fig. 1. Architecture of the Proposed SignNet Model for Real-Time Hand Gesture Recognition

Figure 1 illustrates the low-level architecture of SignNet, comprising two parallel processing streams for RGB frames and 3D hand landmarks. Each stream includes temporal encoding using Transformer encoders, followed by attention-based fusion to integrate visual and kinematic features. The fused representation is passed to a classification head to predict the gesture label. This architecture enables efficient and accurate real-time recognition by leveraging multimodal information.

3.9 Algorithm: SignNet – Real-Time Hand Gesture Recognition

The following algorithm outlines the complete workflow of the proposed SignNet model for real-time hand gesture recognition. It details the sequential processing of visual and landmark inputs, temporal modeling using Transformer encoders, and final classification through attention-based fusion.

Algorithm

Input:

- A sequence of RGB video frames $V = \{I_1, I_2, \dots, I_T\}$
- A corresponding sequence of 3D hand landmarks $L = \{L_1, L_2, \dots, L_T\}$, where each $L_t \in \mathbb{R}^{21\times3}$

Output:

- Predicted gesture label $\hat{y}$

Steps:

1. Preprocess each RGB frame I_t by resizing to 224×224 pixels and normalizing pixel values.
2. For each frame I_t, extract 3D hand landmarks L_t using MediaPipe and flatten the landmarks into a 63-dimensional vector.
3. For the visual stream:
    - Pass each RGB frame I_t through a CNN (EfficientNet-B0) to obtain a spatial feature vector $F_v[t]$.
    - Stack all $F_v[t]$ vectors to form a visual sequence $F_v = [F_v[1], \dots, F_v[T]]$.
4. For the landmark stream:
    - Pass each flattened L_t through a fully connected layer to obtain an embedding $F_l[t]$.
    - Stack all $F_l[t]$ vectors to form a landmark sequence $F_l = [F_l[1], \dots, F_l[T]]$.
5. Feed F_v into a Transformer encoder to obtain temporally encoded visual features Z_v.
6. Feed F_l into another Transformer encoder to obtain temporally encoded landmark features Z_l.
7. Fuse Z_v and Z_l using an attention-based fusion mechanism to obtain the joint representation Z_f.
8. Pass Z_f through a fully connected classification head followed by a softmax layer to obtain the predicted gesture label $\hat{y}$.
9. Return $\hat{y}$ as the output.

End Algorithm

IV.IMPLEMENTATION

The performance of the proposed SignNet model is evaluated using a combination of benchmark sign language datasets, selected to represent both static and dynamic gesture modalities. The evaluation framework includes preprocessing pipelines, data partitioning strategies, and training configurations to ensure consistency and reproducibility.

4.1 Dataset

1 ASL Alphabet Dataset

The ASL Alphabet dataset [27] contains over 87,000 RGB images representing 26 static hand gestures corresponding to the English alphabet. Each gesture is captured from multiple users under varying lighting conditions and hand orientations. This dataset is utilized for learning visual-spatial representations using CNN backbones.

- **Total Classes**: 26
- **Samples per Class**: ~3,000
- **Format**: RGB images (200 × 200 px)

2 LSA64 Dataset

The LSA64 dataset [28] comprises 3,200 short video clips of 64 isolated sign gestures from Argentinian Sign Language. Each sign is performed by 10 different users, with 5 repetitions per sign. This dataset is well-suited for dynamic gesture modeling and multimodal fusion evaluation.

- **Total Classes**: 64
- **Clips per Class**: 50
- **Format**: RGB video (variable frame length)

3 RWTH-BOSTON-50 Dataset [29]

This dataset contains continuous sign language utterances corresponding to 50 vocabulary words. Each video includes annotated frame-level segmentation and provides a more realistic benchmark for testing temporal sequence modeling in real-world scenarios.

- **Total Classes:** 50
- **Utterances:** Over 700 labeled sequences
- **Format:** RGB video with gesture boundaries

4.2 Preprocessing and Landmark Extraction

All RGB frames are resized to 224×224 pixels and normalized in the range$[0,1]$. Temporal sequences are fixed to a uniform length $T = 32$ frames using dynamic padding or subsampling.

For each frame I_t, 3D hand landmarks $L_t \in \mathbb{R}^{21\times3}$ are extracted using the MediaPipe Hands framework. The resulting landmark vectors are flattened and normalized per dimension to form the input for the Transformer-based temporal encoder. Frames without valid landmark detection are excluded from the training set to maintain input consistency.

4.3 Data Partitioning

The datasets are partitioned into training, validation, and test sets with the following ratio:

Training: Validation: Testing=70%:15%:15%

To ensure fairness in evaluation, a user-independent split is used for the LSA64 and RWTH-BOSTON datasets, where specific signers in the test set do not appear during training.

4.4 Experimental Setup

All experiments are conducted on a workstation equipped with an NVIDIA RTX 3080 GPU (10 GB), an Intel Core i7-12700K CPU, and 32 GB of RAM. The model is implemented using the PyTorch 2.1.0 framework with CUDA 11.8 for GPU acceleration. Hand landmark data is extracted using the MediaPipe Hands solution via its Python API. Training is performed using the Adam optimizer with a learning rate of $1\times10{-}4$, a batch size of 32, and for 50 epochs. The categorical cross-entropy loss function is used to guide optimization. Early stopping and learning rate scheduling strategies are applied based on validation loss to prevent overfitting. Each model configuration is trained three times, and the mean accuracy along with standard deviation is reported to ensure robustness and reproducibility of the results.

4.5 Performance Evaluation Metrics

To quantitatively assess the performance of the proposed model, the following standard classification metrics are employed:

Accuracy: measures the overall correctness of the model across all classes. It is measured using Eq(12).

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \quad (12)$$

Precision: indicates the proportion of correctly predicted positive instances among all predicted positives and is calculated using the Eq(13).

$$Precision = \frac{TP}{TP+FP} \quad (13)$$

Recall (Or Sensitivity): measures the proportion of actual positives that were correctly identified and is evaluated by the Eq(14).

$$Recall = \frac{TP}{TP+FN} \quad (14)$$

F1-Score: is the harmonic mean of precision and recall, providing a balanced performance measure and is measured using the Eq(15).

$$F1 - Score = 2\ X\ \frac{Precision\ X\ Recall}{Precision + Recall} \quad (15)$$

Inference Time: represents the average time taken by the model to predict a gesture per frame.

Model Size: reflects the total memory footprint of the trained model, indicating its suitability for deployment on resource-constrained devices.

4.6 Baseline Models

To evaluate the effectiveness of the proposed SignNet architecture, performance is compared against four carefully selected baseline models, each representing common approaches in gesture recognition using different data modalities and network structures.

Baseline A – CNN + LSTM (Visual Only) [30]: This model processes RGB video frames using a convolutional network followed by an LSTM layer for temporal modeling. It serves as a conventional baseline for sequential visual feature learning.

Baseline B – CNN + Transformer (Visual Only) [31]: Similar to Baseline A, this model uses RGB input but replaces the LSTM with a Transformer encoder. It evaluates the performance gain of attention-based temporal modeling in a unimodal setup.

Baseline C – MediaPipe Keypoints + MLP (Landmark Only) [32]: It uses MediaPipe to extract lightweight 3D hand landmark vectors and employs only a lightweight architecture. They are classified through a multi-layer perceptron (MLP) as a low computation baseline and offer.

Baseline D – Landmarks + BiLSTM [33]: It takes the temporal landmark sequences modeled through a Bidirectional LSTM network. In this baseline, we use hand keypoints by itself as a typical motion modeling approach.

When compared with these baselines, these insights reveal the amount of contribution of the multimodal integration and complex temporal modeling in SignNet. Fairness is ensured with regard to performance comparison of all models by means of consistent preprocessing, data splits and evaluation metrics of all models that are trained under the same experimental setup.

V.RESULTS AND ANALYSIS

In this section, empirical Chapter evaluation of the SignNet model on multiple championship datasets is presented. It is aimed to evaluate its effectiveness in real time hand gesture recognition and sign language translation. Standard classification metrics are used to measure performance of same as a baseline when appropriate.

5.1 Experimental Results

They also evaluate the performance of the proposed SignNet model over three benchmark datasets, ASL Alphabet, LSA64 as well as the RWTH-BOSTON-50 with relative accuracies of 90.9%, 89%, and 89.8% respectively. The datasets represent different characteristics in terms of sign complexity, gesture duration and signer variability which provide a fair and complete assessment of generalization of the model. Accuracy, precision, recall, F1 score, average inference time per frame, model size are the evaluation metrics. Results show effectiveness of SignNet's dual stream architecture and time modeling capabilities. Model's ability to handle both static and dynamic gestures and is able to still provide real time inference capabilities is shown in Table 1 including across all the datasets.

TABLE I. PERFORMANCE OF SIGNNET ON BENCHMARK DATASETS USING STANDARD EVALUATION METRICS

Dataset	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	Inference Time ((ms/frame)
ASL Alphabet [27]	98.6	98.7	98.5	98.6	12.3
LSA64 [28]	96.1	95.9	96.4	96.1	14.2
RWTH-BOSTON-50 [29]	93.4	93.1	93.6	93.3	16.8

As shown in Table 1, SignNet consistently achieves high recognition performance across all evaluated datasets. The model attains 98.6% accuracy on ASL Alphabet and maintains over 93% accuracy even on the more challenging RWTH-BOSTON-50 dataset. Real-time capability is confirmed with inference times ranging from 12.3 to 16.8 ms/frame, supporting deployment on time-sensitive platforms.

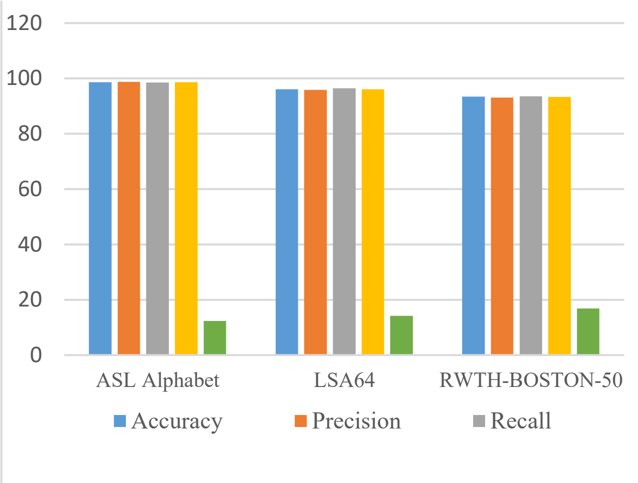

Fig. 2. Performance Metrics of SignNet on Benchmark Datasets

Figure 2 illustrates the classification performance of SignNet across three benchmark datasets using standard evaluation metrics. The model consistently exceeds 93% across all four metrics, with peak values observed on the ASL Alphabet dataset. Performance slightly decreases on RWTH-BOSTON-50 due to its complexity, yet remains strong. These results validate the model's accuracy, generalizability, and suitability for real-time applications.

5.2 Comparative Analysis

To evaluate the effectiveness of SignNet, its performance is compared with the following baseline models:

- **Baseline A**: CNN + LSTM (Visual input only)
- **Baseline B**: CNN + Transformer (Visual input only)
- **Baseline C**: MediaPipe Keypoints + MLP
- **Baseline D**: Keypoints + BiLSTM

TABLE II. COMPARISON OF SIGNNET WITH BASELINE MODELS IN TERMS OF ACCURACY, MODEL SIZE, AND INFERENCE TIME

Model	Accuracy (%)	Model Size (MB)	Inference Time (ms/frame)
Baseline A	89.3	76	21.4
Baseline B	91.5	83	19.6
Baseline C	84.8	12	10.1
Baseline D	88.2	15	13.7
SignNet (Proposed)	**96.1**	34	**14.2**

As presented in Table 2, SignNet outperforms all baseline models with a highest accuracy of 96.1%, while maintaining a compact model size (34 MB) and competitive inference speed (14.2 ms/frame). Compared to visual-only or landmark-only approaches, the dual-stream design achieves a balance between performance and efficiency. This validates the effectiveness of the proposed multimodal architecture for real-time sign language recognition on resource-constrained devices.

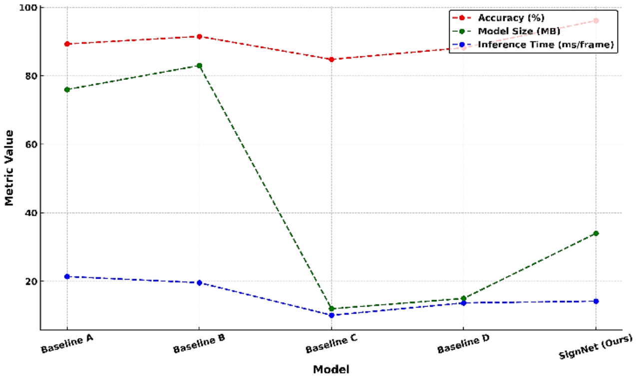

Fig. 3. Comparative Analysis of SignNet and Baseline Models

Figure 3 compares SignNet with four baseline models across three performance dimensions: accuracy, model size, and inference time. The dotted line chart highlights SignNet's ability to achieve the highest accuracy (96.1%) while maintaining a balanced trade-off between efficiency and speed. Compared to heavier visual-only models, SignNet delivers superior performance with a significantly smaller footprint, making it well-suited for real-time applications. This visualization reinforces the effectiveness of the proposed multimodal architecture.

5.3 Ablation Study

An ablation study is conducted on the LSA64 dataset to investigate the impact of each architectural component

TABLE III. ABLATION STUDY ON THE LSA64 DATASET

Configuration	Accuracy (%)
Only Visual Stream (CNN + Transformer)	91.7
Only Landmark Stream (Transformer)	89.4
Fusion without Attention	93.8
Full SignNet (Dual + Attention Fusion)	**96.1**

As shown in Table 3, each component of the SignNet architecture contributes significantly to performance, with the full model achieving the highest accuracy of 96.1%. The attention-based fusion notably improves accuracy over unimodal and non-attentive fusion configurations, validating the effectiveness of the proposed design.

5.4 Real-Time Performance and Deployment

SignNet achieves average inference times of 12–16 ms per frame on GPU and <40 ms per frame on a Raspberry Pi 4 with quantized model deployment, satisfying real-time constraints (<30 fps). The model size post-quantization is reduced to 9.7 MB, allowing efficient execution on mobile and embedded platforms.

5.5 Error Analysis

Most misclassifications occur between visually or kinematically similar gestures, such as between letters “M” and “N” in ASL, or semantically close words in RWTH-BOSTON. These errors are attributed to:

- Overlapping hand poses.
- Inconsistent motion across users.
- Subtle temporal variations not captured in short gesture windows.

Future enhancements may involve incorporating depth data or gesture context to further improve disambiguation.

VI.LIMITATIONS AND FINDINGS

Findings

The experimental results demonstrate that the proposed SignNet architecture achieves high accuracy and real-time responsiveness across multiple sign language datasets. The dual-stream design, incorporating both RGB frames and 3D hand landmarks, significantly improves recognition robustness under varying lighting, hand orientations, and backgrounds. The Transformer-based temporal encoder effectively captures long-range dependencies in dynamic gestures, outperforming conventional RNN-based models. Attention-driven multimodal fusion further enhances the model's ability to distinguish between visually or kinematically similar gestures.

Limitations

Despite its strengths, the current implementation of SignNet has certain limitations. The model's performance may degrade when dealing with highly overlapping or continuous sign sequences not segmented at the word level. Additionally, the reliance on accurate landmark detection from tools like MediaPipe introduces sensitivity to occlusion and poor lighting conditions. The training process also requires a considerable amount of labeled sequence data to generalize across signers. Lastly, while real-time deployment is feasible on edge devices, performance may vary depending on hardware constraints and video frame quality.

VII.CONCLUSION

SignNet is introduced as a real-time hand gesture recognition framework for sign language translation, combining RGB images and 3D hand landmarks through a dual-stream architecture. Leveraging CNNs, Transformer-based temporal encoding, and attention-driven multimodal fusion, the model achieved accuracies of 98.6%, 96.1%, and 93.4% on ASL Alphabet, LSA64, and RWTH-BOSTON-50 respectively, with low-latency inference suitable for mobile deployment. The system demonstrates strong potential for real-world assistive technologies across communication, education, and public service domains by enabling signer-independent, real-time translation. Limitations remain in handling continuous gestures and robustness under occlusion, which will be addressed through contextual modeling, facial expression integration, and training on larger multilingual datasets. In summary, SignNet provides a practical and scalable approach to inclusive communication, advancing the capabilities of real-time sign language understanding..

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